MILITARY SPECIFICATION
FOR
CASE, CARTRIDGE, 105MM, M115

1. SCOPE

1.1 This specification covers Case, Cartridge, 105mm, M115 for Cartridge APDS-T, M392 for M68 Gun.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitations for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-G-2550	- General Specification for Ammunition, Except Small Arms Ammunition
MIL-P-10025	- Packing and Marking for Domestic Shipment of Inert Ammunition Components, General Specification for

STANDARDS

Federal

Federal Test Method Standard No. 151 - Metals; Test Methods

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Inspection Terms and Definitions
MIL-STD-639	- Surface Roughness Values Preferred for Army Ordnance Material

FSC 1315

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DRAWINGS

Ordnance	
F-8597490	- Case, Cartridge, 105MM, M115
F-7548320	- Cartons, Ammunition for Interplant Shipment of Brass Cartridge Cases, Empty, 75MM to 155MM, Inclusive
IEL-8597490	- Inspection Equipment List for Case, Cartridge, 105MM, M115

PUBLICATIONS

ORD-M608-11	- Ordnance Inspection Handbook - Procedures and Tables for Continuous Sampling by Attributes

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. - The cartridge case shall comply with the requirements of the applicable drawings, referenced specifications and the following:

3.2 Preproduction Sample. - Unless otherwise directed by the contracting officer, a preproduction sample shall be required.

3.3 Material. - The contractor shall be responsible for selecting from the classification list contained in the raw material specification, the surface and physical condition of the raw material intended for use in manufacture.

3.4 Annealing. - Annealing temperature shall not exceed 1200 degrees Fahrenheit (F).

3.5 Stress Relief. - After the last cold forming operation, the cartridge case shall be given a stress relief treatment of approximately 500 degrees Fahrenheit (F) for one hour.

3.6 Mouth Anneal. - After the final forming operation, the cartridge case shall be annealed at the mouth for a distance of 4 inches minimum, 4 1/2 inches maximum.

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3.7 Mercurous Nitrate. - The cartridge case when fitted with a steel mouth plug simulating the maximum dimensions of the 105MM, M392 projectile, shall show no evidence of cracking after immersion in mercurous nitrate.

3.8 Maximum Profile. - The cartridge case shall freely enter the Ordnance certified maximum profile gage in such a manner as to permit the slide to close without interference.

3.9 Mechanical Properties.

3.9.1 Hardness. - The metal .120 inch minimum, from the surface around the primer hole shall have a Rockwell hardness value of not less than B75

3.9.2 Tensile Strength. - Specimens taken from sample cartridge cases in accordance with Figure I shall comply with the requirements of Table I. (See 6.1).

TABLE I

	Pounds per Square Inch (PSI)
Specimen A	44,000 Minimum - 57,000 maximum
Specimen B	50,000 Minimum
Specimen C	80,000 Minimum
Specimen D	75,000 Minimum
Specimen E	75,000 Minimum

Elongation shall be recorded for information only.

3.10 Functional. - The cartridge case shall be subjected to a functional test in the following phases:

a. Excess Pressure Phase
b. Excess, Excess Pressure Phase (for information only)

3.10.1 Obturation. - The cartridge case shall effectively seal the breech of the weapon against leakage of hot gases generated during firing.

3.10.2 Soundness. - After firing, the cartridge case shall be free of cracks, except as permitted by Table III.

3.10.3 Ejection. - When fired, the cartridge case shall be ejected by the service extractors when the weapon is operated in the normal manner. (See 6.2).

3.10.4 Primer Retention. - The cartridge case shall retain the primer during firing and ejection.

3.11 Workmanship. - The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

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3.11.1 Metal Defects. - The cartridge case shall be free from cracks, blisters, seams, laminations, deep scratches and porosity.

3.11.1.1 Folds and Cold Shuts. - The cartridge case shall contain no folds or cold shuts in the radius area extending beyond the tangents from the base and sidewall, however, at no time will any fold or cold shut exceed .035 maximum length. See Figure II.

3.11.2 Surface Finish. - The requirements for surface finish are as detailed on the applicable drawing.

3.11.3 Threads. - Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawing and provisions of Specification MIL-G-2550.

3.11.4 Burr. - No cartridge case shall have a burr which might interfere with the assembly of the primer or projectile, or which might be injurious to personnel handling the cartridge case.

3.11.5 Foreign Matter. - No cartridge case shall contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 General Quality Assurance Provisions. - The quality assurance provisions in this section and other documents referenced in this specification are applicable to inspection performed by or for the Government. Reference shall be made to Standard MIL-STD-109 to define the inspection terms used. Inspection shall be in accordance with Specification MIL-G-2550 except as specified herein.

4.1.1 Contractor Inspection. - Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.2 Contractor Quality Assurance System. - The contractor shall provide and maintain an adequate quality assurance system, acceptable to the Government, covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by

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this specification and other applicable documents referenced in this specification. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of assurance required by this specification or other applicable documents referenced herein.

4.1.3 Government Verification. - All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviations from the prescribed or agreed upon procedures or instances of poor practices which might have a detrimental effect upon the quality of the product will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered will be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor shall designate a responsible official or officials to whom the Government inspector will report such instances.

4.2 Preproduction Inspection. - Prior to commencing regular production, the contractor shall submit a preproduction sample to a designated Government approved facility for evaluation. The sample shall consist of seventy (70) cartridge cases. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the preproduction sample shall be inspected for all requirements of the drawing and specifications including all those characteristics listed in 4.3.3 and the tests in 4.3.4. The Government reserves the right to require new preproduction samples until such time as an acceptable sample is submitted. Sixty (60) of the above cartridge cases shall be forwarded to a designated proving ground for ballistic testing.

4.3 Inspection Provisions. - Alternative inspection procedures and inspection equipment may be used by the contractor when such procedures and equipment provide as a minimum, quality assurance required in the contractual documents. The inspection procedures or equipment specified in this specification will prevail in case of dispute.

4.3.1 Material. - Reports of objective tests performed on the material by the contractor shall be required to insure compliance with the chemical and physical requirements. Reports shall include positive identification

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of the material and part, the applicable requirements, the specific tests performed and the results obtained. Numerical results of each specimen or sample inspected shall be included except that where numerical results are impossible an adjective description shall be used. Initial test reports shall accompany the preproduction sample. Prior to inspection, the inspector shall determine from the information stated on test reports whether the material submitted complies with the requirements. Components made from material which does not comply shall be rejected.

4.3.2 Lot.

4.3.2.1 Submittal of Product. - The contractor shall submit the product in accordance with Standard MIL-STD-105 or ORD-M608-11 as applicable.

4.3.2.2 Size of Lot. - For the purposes of identification and shipment, the lot size shall be as agreed upon between the contractor and the contracting officer, provided the homogeneity of the inspection lot has been retained.

4.3.3 Examination. - Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105 or ORD-M608-11 as applicable. An AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect shall be used. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter. The Ordnance standard finish blocks prescribed in Standard MIL-STD-639 shall be used as a basis of comparison for surface roughness determination.

4.3.3.1 Maximum Profile Gage. - Each cartridge case shall be gaged in an Ordnance certified maximum profile gage. Failure of any cartridge case to comply with requirements of 3.8 shall be cause for rejection of the cartridge case.

4.3.3.2 Classification of Defects.

4.3.3.2.1 Case, Cartridge, Drawing F-8597490.

	Defect	Method of Inspection
MAJOR		
1.	Pitch diameter of primer thread, max.	Gage
2.	Minor diameter of primer thread, max.	Gage
3.	Concentricity of primer hole thread with body	Gage
4.	Concentricity of primer hole counterbore with thread	Gage
5.	Concentricity of primer hole thread relief with thread	Gage

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Defect	Method of Inspection
MAJOR (Cont'd)	
6. Diameter of flange, min.	Gage
7. Diameter of mouth	Gage
8. Length of mouth diameter, min.	Gage
9. Thickness of flange, min.	Gage
10. Wall thickness at datum near head	Gage
11. Wall thickness at datum near mouth	Gage
12. Wall thickness at mouth	Gage
13. Thickness of head, min.	Gage
14. Depth of primer hole counterbore	Gage
15. Depth of primer hole thread relief	Gage
16. Metal defective	Visual
17. Contour of inside of head and sidewall not smooth	Visual
MINOR	
41. Pitch diameter of primer thread, min.	Gage
42. Diameter of primer hole counterbore	Gage
43. Diameter of primer hole thread relief	Gage
44. Diameter of body under flange	Gage
45. Length to first datum diameter from head end	Gage
46. Length to second datum diameter from head end	Gage
47. Length to third datum diameter from head end	Gage
48. Length to fourth datum diameter from head end	Gage
49. Total length	Gage
50. Chamfer on head missing or incorrect	Visual
51. Surface finish, improper	Visual
52. Burr	Visual
53. Marking missing, incorrect or illegible	Visual
54. Foreign matter	Visual

4.3.3.2.2 Packed Carton, Prior to Sealing, Drawing F-7548320.

MAJOR	
None.	
MINOR	
41. Support missing or damaged	Visual
42. Separator missing or damaged	Visual
43. Improper assembly	Visual
44. Incorrect number in carton	Visual

4.3.3.2.3 Packed Carton, After Sealing, Drawing F-7548320.

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Defect	Method of Inspection
MAJOR	
None	
MINOR	
41. Carton damaged, cut or gouged through all layers	Visual
42. Carton not completely sealed	Visual
43. Contents loose	Manual/Visual
44. Marking missing, incorrect or illegible	Visual

4.3.4 Inspection Testing. - The following tests shall be performed in accordance with the provisions of 4.4, except as provided in 4.3.

4.3.4.1 Mercurous Nitrate. - Five cartridge cases from the preproduction sample and five cartridge cases from each production lot shall be subjected to mercurous nitrate immersion. Failure of one cartridge case to comply with the requirements of 3.7 shall be cause for rejection of the sample or lot.

4.3.4.2 Mechanical Properties.

4.3.4.2.1 Hardness. - Five cartridge cases from the preproduction sample and five cartridge cases from each production lot shall be subjected to this test. Failure of one cartridge case to comply with the requirements of 3.9.1 shall be cause for rejection of the sample or lot.

4.3.4.2.2 Tensile Strength. - Five cartridge cases from the preproduction sample and five cartridge cases from each production lot shall be subjected to this test. Failure of one cartridge case to comply with the requirements of 3.9.2 shall be cause for rejection of the sample or lot.

4.3.4.2.3 Elongation. - The percentage of elongation of each specimen used for tensile strength test shall be measured and recorded for information only.

4.3.4.2.4 Macro Examination. - Five cartridge cases from the preproduction sample and five cartridge cases from each production lot shall be examined for folds and cold shuts. Failure of one cartridge case to comply with the requirements of 3.11.1 will be cause for rejection of the sample or lot.

4.3.4.3 Functional. - The following functional tests will be performed by the Government.

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4.3.4.3.1 Preproduction Sample. - The quantities listed in Table II will be ballistically tested in the phases indicated and the results of the test will be evaluated in accordance with criteria set forth in Table III except that no retest will be permitted.

TABLE II

Test Phase	Quantity to be Tested
Excess Pressure Phase	30
Excess, Excess Pressure Phase (for information only)	30

4.3.4.3.2 Production Lot Sample. - Thirty cartridge cases from each production lot of 25,000 or less will be ballistically tested in the excess pressure phase. Lots which exceed 25,000 cartridge cases, one additional cartridge case will be tested for every one thousand cases (or fraction thereof) above 25,000. Acceptance and rejection criteria shall be in accordance with Table III. The lot shall be considered eligible for retest using double the number of samples provided that after firing, no cartridge case contains crack(s) or split(s) within one caliber of the head, and the rejection criteria of Table III has not been attained.

TABLE III

Defect Characteristic	Accept	Retest	Cumulative Rejects
1. Obturation	0	1	2
2. Splits or Cracks within 1 caliber of Head	0	0	1
3. Splits or Cracks in Sidewall more than 1 caliber from Head	1	2	3
4. Ejection	0	1	2
5. Primer Retention	0	1	2

4.3.5 Process Control. - Monitoring at unscheduled intervals shall be conducted for the following process operations to determine compliance with following requirements.

Anneal - 3.4
Stress Relief - 3.5
Mouth Anneal - 3.6

Failure to comply with the requirements shall be cause for rejection of all units processed since the last monitored approval of the process.

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4.3.6 Inspection Equipment. - The examination and tests shall be made using equipment listed on IEL-8597490, except as provided in 4.3. The Government will determine by means of its monitoring program, that the contractor has available and correctly utilizes gaging, measuring and test equipment of required accuracy and precision and that the instruments are of proper type and range to make measurements of the required accuracy. The contractor shall have available a set of master gages, standards and other appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration shall be maintained by the contractor and be made available to the Government for review. The calibration of gages, standards and instruments will be periodically checked by authorized Government personnel.

4.4 Test Methods and Procedures.

4.4.1 Mercurous Nitrate. - The cartridge case, fitted with a steel mouth plug in accordance with Figure III, shall be tested in accordance with the procedures prescribed by Method 831, Federal Test Method Standard No. 151.

4.4.2 Hardness. - The cartridge case shall be sectionalized through the center of the head. Three (3) or more hardness readings each shall be taken equally spaced through the head on both sides of the primer hole with the center line of the hardness indents located 0.100 inch from the major diameter of the primer hole thread. The hardness readings shall be made in accordance with the procedures prescribed by Method 243.1, Federal Test Method Standard No. 151.

4.4.3 Tensile Strength and Elongation. - The test specimens shall be taken from the cartridge case in accordance with the standard positions and size of tensile specimen chart, Figure I. The tensile strength and elongation shall be determined in accordance with the procedures for these tests prescribed by Method 211.1, Federal Test Method Standard No. 151.

4.4.4 Macro-Examination. - The cartridge case shall be sectionalized through the center of the head, polished and examined for folds and cold shuts in the radius areas of the head in accordance with the procedures prescribed by Method 321.1, Federal Test Method Standard No. 151. One hundred (100X) magnification or more shall be used for this examination.

4.4.5 Functional.

4.4.5.1 Preproduction Sample.

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4.4.5.1.1 Excess Pressure Phase. - The cartridge case will be assembled into a complete round in the same manner as required for service firing. Complete round test components simulating the standard components may be used in lieu of the standard components. The propellant charge will be such as to produce the chamber pressure specified on the applicable drawing. The complete round will be temperature conditioned at plus 65 degrees F to plus 75 degrees F and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4.4.5.1.2 Excess, Excess Pressure Phase. - The cartridge case will be assembled into a complete round in the same manner as required for service firing. Complete round test components simulating the standard components may be used in lieu of the standard components. The propellant charge will be such as to produce the maximum chamber pressure permitted by the proving ground. The complete round will be temperature conditioned at plus 65 degrees F to plus 75 degrees F and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4.4.5.2 Production Sample.

4.4.5.2.1 Excess Pressure Phase. - The cartridge case will be assembled into a complete round in the same manner as required for service firing. Complete round test components simulating the standard components may be used in lieu of the standard components. The propellant charge will be such as to produce the chamber pressure specified on the applicable drawing. The complete round will be temperature conditioned at plus 65 degrees F to plus 75 degrees F and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level A - Not applicable.

5.1.2 Level B - Not applicable.

5.1.3 Level C - Cartridge cases shall be packed in accordance with requirements of Drawing F-7548320.

5.2 Labeling and Marking. - Labeling and marking for shipment shall be in accordance with MIL-P-10025.

5.3 Data Card. - Data card information shall be as specified in Specification MIL-C-2550.

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6. NOTES

6.1 It is not implied that cartridge cases that contain minimum tensile strength values for all specimens will function satisfactorily in ballistic test.

6.2 The term "extraction" shall mean that the fired cartridge case is loosened and pulled to the rear of the weapon sufficiently to permit removal by hand.

6.3 The term "ejected" shall mean that the cartridge case is thrown free of the breech of the weapon to permit loading of the subsequent round.

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CUSTODIAN:

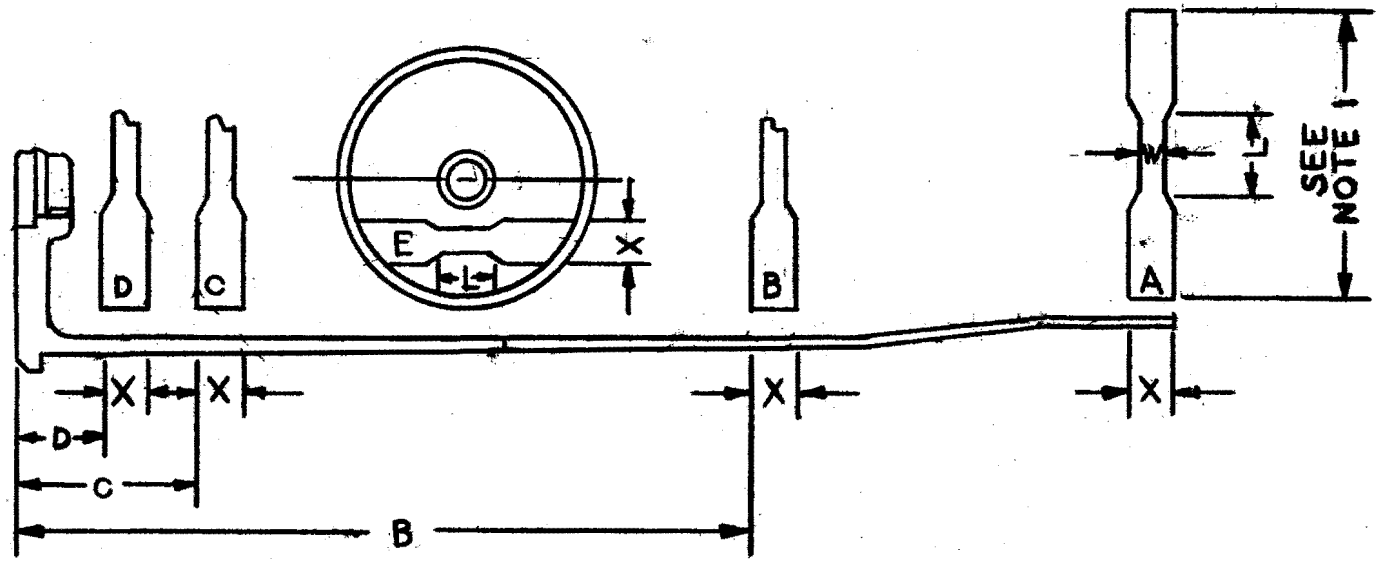
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PREPARING ACTIVITY:

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STANDARD POSITIONS AND SIZES OF TENSILE SPECIMENS FOR CASE, CARTRIDGE, 105MM, M115



SPECIMEN	X	W	L	LOCATION FROM BASE
A	1.25	1.0	2.0	At Mouth
B	1.25	1.0	2.0	13.50
C	1.25	1.0	2.0	2.75
D	1.25	1.0	2.0	0.75
E	0.4	0.3	1.0	In Base

NOTES:

1 - Overall length of specimen to suit fixture in testing machine.

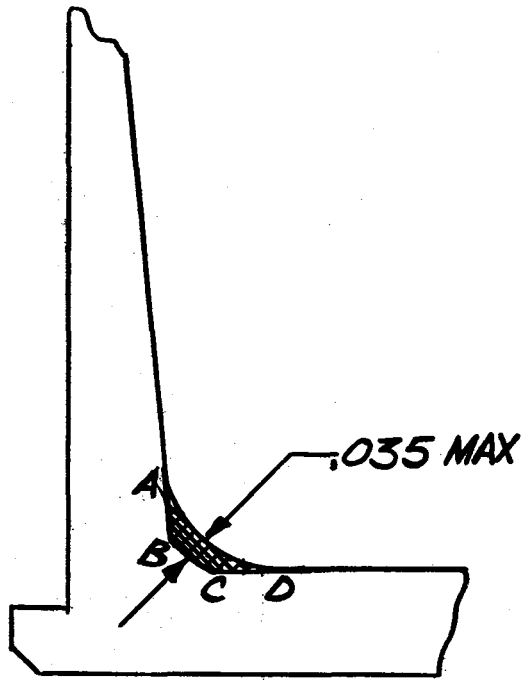
2 - Specimen E to be machined so as to eliminate effects of stamping on test.

3 - Specimens C and D to be machined to uniform thickness.

4 - Centerline of specimen E to be 1.75 inches from centerline of cartridge case.

FIGURE I

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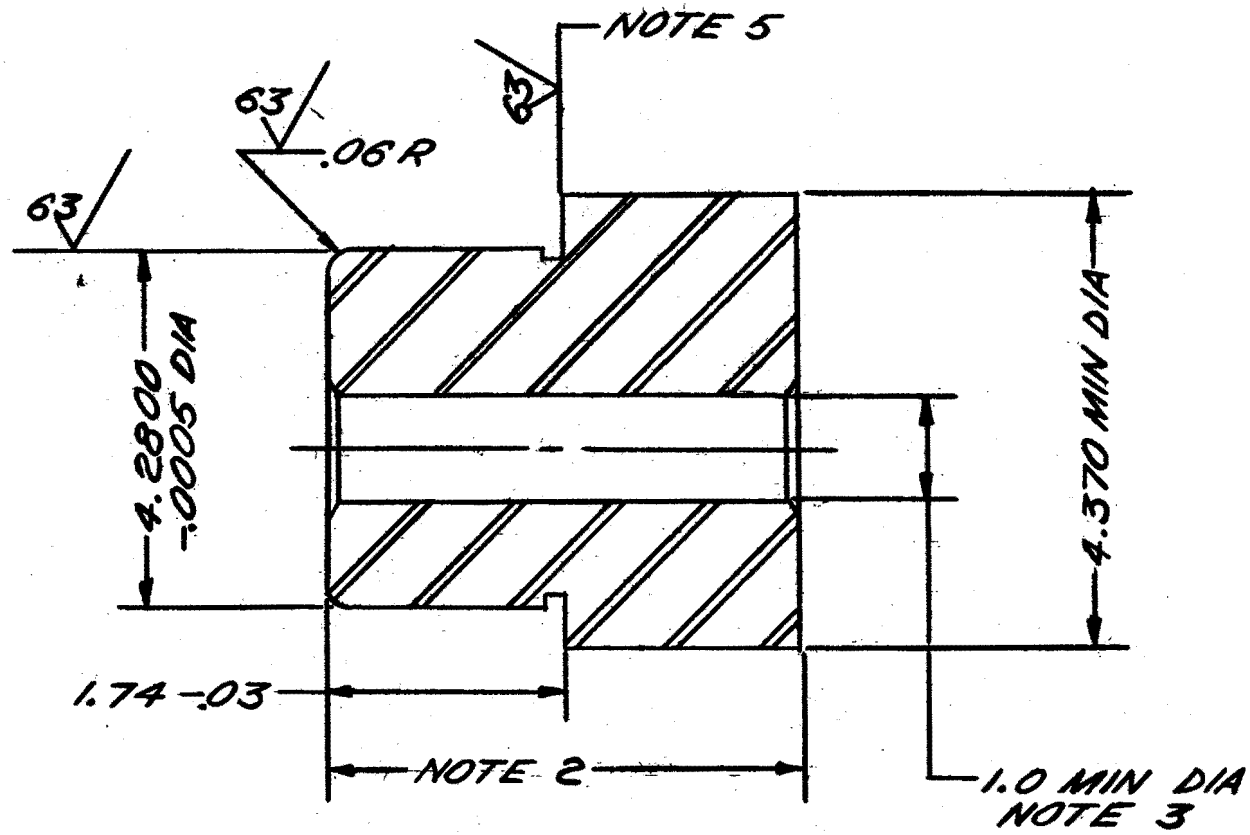


MACRO EXAMINATION

Folds or Cold Shuts shall not extend beyond shaded area inclosed by tangent AB, tangent CD and Arc BC.

FIGURE II

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NOTES:

1 - Material: - Steel
2 - Overall length of specimen to suit testing procedure
3 - Centers permitted; hole may be threaded, for extracting plug
4 - 250 all over except as noted
5 - Cartridge case to be seated against this surface

MOUTH PLUG FOR USE IN MERCUROUS NITRATE TEST

FIGURE III

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